SC-TPU + RV32I SoC on the Tang Nano 9K

Engineering Report — design revision C

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Evidence status. Every result in this report is SIMULATION (Python golden model, RTL simulation with Icarus Verilog, or synthesis / place-and-route / static-timing estimates of the open-source flow), CALCULATION, SOURCE or ASSUMPTION. **No hardware measurement exists.** A bitstream was generated as a proof of routability and was **never programmed** into a device. Simulation is not test; passing synthesis and place-and-route is not hardware validation. Section 11 plans the physical validation that remains to be done.

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1 Scope, conventions and summary

1.1 System

The project delivers a system-on-chip for the Sipeed Tang Nano 9K board (Gowin GW1NR-LV9QN88PC6/I5): a multi-cycle RV32I CPU, 4 KiB of RAM, a peripheral block (6 LEDs, UART transmitter, cycle counter, test mailbox) and a stochastic-computing tensor processing unit (SC-TPU) that runs a 64-16-10 multilayer perceptron for the 8×8 handwritten-digits set. The SC-TPU is a 16×8 weight-stationary tile of bipolar XNOR multipliers with accumulative parallel counters (APC), driven by 8-step leap-forward LFSRs. Its behaviour is specified bit-exactly so that a Python golden model and the RTL agree bit for bit. There is no firmware: CPU programs are testbench stimuli generated by a Python assembler.

1.2 Conventions

- Number labels: SOURCE (data sheet, standard, paper; listed in `management/SOURCES.md`), CALCULATION (formula and inputs shown), SIMULATION (tool, version, log path), ASSUMPTION (reason given). MEASUREMENT is reserved for real hardware and does not occur.
- Status words: **PASS**, **PARTIAL**, **FAIL**, **BLOCKED**, **NOT RUN**. Requirement verification statuses are assigned only in the status matrix after the independent review (gate G4); this report does not pre-empt them.
- Every number in the LaTeX source has a `% evidence:` comment with the file it was taken from. Paths are relative to the repository root.
- Design revisions: rev A (first frozen spec), rev B (seed change, superseded, negative result), rev C (current; `design/architecture/ARCHITECTURE.md` sec. 12, `management/INTERFACES.json` revision C).

1.3 Key results

2 Requirements and verification status

The requirement baseline is `requirements/REQUIREMENTS.md` issue 2 (task REQ-02, spec rev C): 82 requirements in eight areas (SYS 17, CPU 10, TPU 16, AI 11, RES 6, TIM 5, DOC 8, VER 9), each with text, rationale, source, verification method (A analysis, S simulation, I inspection, T test on hardware) and acceptance criterion. Method T is never executed in this project; every T item belongs to the physical validation plan of Section 11. Numeric criteria are backed by `requirements/BUDGETS.md` issue 2.

2.1 Status matrix

Requirement status matrix: pending G4.

The matrix (`requirements/STATUS_MATRIX.md`) is written by the systems-engineer after the independent review REV-01. Until then every one of the 82 requirements keeps the status NOT RUN of `requirements/REQUIREMENTS.md` issue 2; the evidence overview in Table 2 is *not* a verification status.

2.2 Evidence overview (not a status)

Table 2 lists, for the requirements with numeric acceptance criteria, the evidence that the specialists reported. It is an aid for the reviewer and the systems-engineer; the formal status is set in the matrix above.

Table 1: Headline results of design rev C (all SIMULATION or CALCULATION; none is a hardware measurement).

Quantity	Value	Label	Evidence
SC accuracy, mode 2, $K = 8$, deployed seeds	94.4 % (340/360)	SIMULATION	`reports/sw/accuracy_revC.json`; Wilson CI 91.6–96.4 %
Expected-value (EV) and int8 reference	95.0 % (342/360)	SIMULATION	same
Seed-ensemble mean, mode 2, $K = 8$ (20 offsets)	92.8 ± 1.4 %	SIMULATION	same
RTL vs golden model, rev C	0 mismatching fields / 1454 inferences	SIMULATION	`logs/tpu/sim_tpu_console.log`
CPU vs ISS	176/176 runs (44 programs × 4 bus timings)	SIMULATION	`logs/cpu/sim_cpu_summary.log`
SoC end-to-end demo	RESULT 7, LED, UART "7\n", TOHOST 1	SIMULATION	`logs/soc/tb_soc_demo.log`
SoC LUT4 / DFF / BSRAM	5628 (65 %) / 2803 (43 %) / 5 (19 %)	SIMULATION	`reports/fpga/soc/soc_impl.md`
SoC Fmax, worst of 3 seeds	47.46 MHz (target ≥ 29.7)	SIMULATION	same (STA, not silicon-validated)
TPU latency, $K = 8$	2928 cycles = 108.4 µs	SIM. + CALC.	`logs/tpu/tb_sc_tpu_mode2.log` D8
Bitstream	3 556 680 bytes, never programmed	file property	`logs/soc/pack.log`

Table 2: Specialist-reported evidence for requirements with numeric acceptance (SIMULATION / CALCULATION; not a verification status).

ID	Acceptance (short)	Reported evidence	Evidence file
REQ-SYS-005	rst latency 2–3 edges; first fetch ≤ 40 cycles	2 edges; first fetch 31 cycles after rst falls	`logs/soc/tb_soc_demo.log`, RTL-SOC R1–R5
REQ-SYS-007	0 protocol violations; slave ≤ 4 cycles (TPU buffer read while BUSY ≤ 40, ADR-012)	0 violations; max 3 cycles in demo; 35 cycles worst TPU stalled read	RTL-SOC, RTL-TPU handoffs
REQ-SYS-012	|baud error| ≤ 2.0 %; 234 clk/bit; frame 2340	+0.16 % (CALCULATION); 234 / 2340 clocks; 0x00 0x55 0xA5 0xFF decoded	`logs/soc/tb_soc_bfm.log`, `soc_periph.v`
REQ-SYS-016	TOHOST=1 within 200 000 cycles; RESULT = golden class	cycle 5441; RESULT 7 = golden	`logs/soc/tb_soc_demo.log`
REQ-CPU-010	random programs match ISS	20 random programs × 4 timings PASS	`logs/cpu/sim_cpu_summary.log`
REQ-TPU-003	PARAMS 0x0302_0810 (mode 2)	0x03020810 read	`logs/tpu/sim_tpu_console.log` D2
REQ-TPU-013	0 mismatches vs rev C golden model	1454 inferences + 239 clamp re-runs, 0 fields	`logs/tpu/tb_sc_tpu_mode{0,1,2}.log`
REQ-TPU-014	CYCLES = BUSY edges	= formula $10 \cdot 2^K + 368$, every vector	same
REQ-AI-003	float, EV ≥ 94.0 %	95.0 % / 95.0 %	`reports/sw/accuracy_revC.json`
REQ-AI-004	≥ 90.0 % (≥ 324/360) at $K = 8$	94.4 % (340/360)	same
REQ-AI-005	$K_{op} \le 9$, gap ≤ 2 pp for $K \ge K_{op}$	$K_{op} = 8$, gaps 0.56 / 0.56 / −0.28 pp	same, `requirement_checks`
REQ-AI-006	ensemble mean ≥ 90 %; |deployed−mean| ≤ 2σ	92.8 %; 1.14 σ	same
REQ-RES-001..004	each resource ≤ 85 %	worst 65 % (LUT4); BSRAM 5/26; DSP 0	`reports/fpga/soc/soc_impl.md`
REQ-RES-005	10 I/O	10	`logs/soc/pins.log`

ID	Acceptance (short)	Reported evidence	Evidence file
REQ-TIM-001	worst-seed Fmax $\geq$ 29.7 MHz	47.46 MHz	`logs/soc/pnr_nwl_s{1,2,3}.log`
REQ-TIM-004	CYCLES(K) $\leq$ 1.10 $N_{budget}(K)$; $K=8$: 3265	2928 at $K=8$; all K below bound	`requirements/BUDGETS.md` sec. 3
REQ-TIM-005	CPU-side time reported	only START$\rightarrow$RESULT 2948 and reset$\rightarrow$TOHOST 5441 (inputs not written by the demo)	`logs/soc/tb_soc_demo.log`, ADR-011
REQ-VER-005	synth, P&R, pack exit 0	all exit 0; bitstream sha256 recorded	`logs/soc/pack.log`
REQ-VER-009	physical validation plan present	Section 11 (all steps NOT RUN)	this report

3 Architecture and decisions

3.1 Block structure

- **Clock and reset.** One 27 MHz domain from the board oscillator (pin 52), no PLL. Button S1 (assumed pin 3) passes a 2-flip-flop synchroniser into a synchronous active-high reset; there is no debouncer.
- **CPU** (`design/rtl/cpu/`). RV32I multi-cycle FSM (CLEAR, FETCH, DECODE, EXECUTE, SHIFT/MEM, WB), register file in two BSRAM blocks cleared by a 32-cycle CLEAR state, one 33-bit adder, iterative shifter, no CSRs, no interrupts; ECALL, EBREAK, illegal and misaligned events halt the core with a trap cause.
- **Bus** (`design/rtl/soc/soc_bus.v`). Native valid/ready, one master; decoder on `addr[31:28]`; AND-OR read multiplexer; unmapped responder (read 0, sticky `bus_err`).
- **RAM** 4 KiB BSRAM with byte enables, initialised from the demo program; **peripherals**: LED, UART TX (234 clocks/bit), CYCLE, TOHOST.
- **SC-TPU** (`design/rtl/tpu/`): bus slave and register map (`sc_tpu.v`), controller FSM (IDLE, LOAD, STREAM, POST) (`sc_tpu_ctrl.v`), RNG modes (`sc_rng.v`), 8-step LFSRs with look-ahead register (`sc_lfsr8.v`), lane array with 8 input SNGs, 128 weight SNGs, XNOR, popcount and 16 17-bit APC accumulators (`sc_lane_array.v`), two-stage post-processing with a 34-bit exact shifter (`sc_postproc.v`); weight RAM 1 BSRAM, INPUT/BIAS in LUT-RAM.

3.2 Algorithm of one inference

For layer 1 (64 inputs, 8 tiles) and layer 2 (16 inputs, 2 tiles): clear the accumulators; per tile load 128 weights and 8 input codes, then run $L = 2^K$ stream cycles, each adding $\sum_i [r_x < x_i] \oplus [r_w < W_{ji}]$ to acc_j; post-process $D = 2\,acc - N_{in} L$, $B = \lfloor bias\, 2^K/128 \rfloor$, $D' = D + B$; layer 1 writes HIDDEN = 128+clip($\lfloor D'\, 2^{s+7}/2^K \rfloor, 0, 127$), layer 2 writes SCORE = D'; RESULT is the argmax over lanes 0..9 (ties to the lowest index). The LFSRs are re-seeded on START and advance only in stream cycles.

3.3 Architecture decision records

Table 3: Summary of the architecture decision records (`management/DECISIONS.md`).

ADR	Subject	Decision	Evidence / consequence
000	Specialist execution mode	Specialists first ran as general-purpose subagents obeying their role files; native agent types after a restart (addendum)	process deviation, labelled
001	Accelerator arithmetic	Bipolar SC with XNOR + APC; int8 binary kept as analytical baseline; unipolar SC rejected	first area estimate ($\sim$9 LUT/lane) corrected by ADR-007
002	Dataflow	16$\times$8 weight-stationary tile; IL studied as 4/8/16 variant	fully parallel layer needs 8192 FF (>6480)
003	RNG decorrelation	Modes 0/1/2 as synthesis parameter, selection by accuracy vs cost	closed by ADR-008/010 (mode 2)

ADR	Subject	Decision	Evidence / consequence
004	CPU micro-architecture	RV32I multi-cycle (most area left for the TPU)	CPI 4.231 at 0 wait states
005	Leap-forward LFSR	8 steps per stream cycle (8 fresh bits per r)	8 LUT4 and one level per LFSR
006	Spec rev A.1	Misaligned jump traps on the jump; strict illegal decoding; reset values	applied in CPU/TPU RTL
007	Spec rev A.2, budget corrections	CONFIG clamp read-back, CYCLES definition, reserved offsets, reads while BUSY, BIAS sign extension, widths, margin definition ($\leq$85 %, $\geq$29.7 MHz); tile cost 2855 LUT4 / 1454 FF	requirements re-based
008	Rev B	RNG_MODE 2 default; mode-2 seeds maximally phase-spaced	**found defective** (ADR-009)
009	Rev B defect; rev C protocol	Translated operand-pair streams; seed rule; pre-registered selection on validation data	SW-02, SW-03
010	Rev C adopted	C0 = rev A mode-2 seeds; modes 0/1 keep 0xACE1/0xF9DE; R1+R2 documented; PARAMS 0x03	closes ADR-003/008/009
011	Requirement adjustments	REQ-SYS-005(b) bound 8 $\rightarrow$ 40 cycles; REQ-TIM-005 PARTIAL accepted for V1; release flow `-nowidelut`	RTL-SOC evidence
012	REQ-02 rulings	TPU buffer reads while loading may take $\leq$40 cycles; accuracy thresholds confirmed; vendor-host access remains a user decision	applied by the systems-engineer

4 Interfaces

4.1 Memory map

Table 4: Memory map (SOURCE: ARCHITECTURE.md sec. 4, INTERFACES.json revision C).

Base	Size	Slave	Notes
`0x0000_0000`	4 KiB	RAM (BSRAM)	program + data, reset PC 0; aliased in `0x0xxx_xxxx`
`0x1000_0000`	0x20	peripherals	Table 5; offsets 0x14.. read 0
`0x2000_0000`	4 KiB	SC-TPU	Table 6
other	–	none	read 0, write ignored, `mem_ready` returned, sticky `bus_err`

4.2 Register maps

Table 5: Peripheral registers, base `0x1000_0000` (SOURCE).

Offset	Name	Access	Content
0x00	LED	RW	[5:0], 1 = LED lit (pins are active-low, the top inverts)
0x04	UART_TX	W	[7:0] byte to send; ignored while busy
0x08	UART_STATUS	R	[0] tx_busy (whole frame incl. stop bit)
0x0C	CYCLE	R	free-running 32-bit cycle counter since reset
0x10	TOHOST	RW	testbench mailbox: 1 = pass, other odd value = fail code

Writes to buffers, weights, CONFIG or CTRL while BUSY are ignored; reads while BUSY return the current storage contents and may be stalled while the controller owns the memory ports.

4.3 Bus protocol

The master holds `mem_valid`, `mem_addr` (32), `mem_wdata` (32), `mem_wstrb` (4; 0000 = read) and `mem_instr` stable until the slave asserts a one-cycle `mem_ready`; `mem_rdata` is valid in that cycle; slaves never assert

Table 6: SC-TPU registers, base `0x2000_0000` (SOURCE; rev C).

Offset	Name	Access	Content
0x000	CTRL	W	[0] START (ignored while BUSY)
0x004	STATUS	R	[0] BUSY, [1] DONE (cleared by START)
0x008	CONFIG	RW	[3:0] LOG2_L (reset 8, clamped to 4..10, reads back the clamped value), [6:4] ACT_SHIFT (reset 0)
0x00C	RESULT	R	[3:0] argmax class
0x010	CYCLES	R	rising edges with BUSY = 1 of the last inference
0x014	PARAMS	R	[7:0] NL = 16, [15:8] IL = 8, [17:16] RNG_MODE, [31:24] version 0x03; release build reads 0x0302_0810
0x040–0x07F	INPUT[0..63]	RW	one code per byte, 4 per word, little-endian
0x080–0x0BF	SCORE[0..15]	R	signed 32-bit per output lane
0x0C0–0x0CF	HIDDEN[0..15]	R	one code per byte
0x100–0x17F	BIAS[0..31]	RW	[15:0] signed, read back sign-extended; word $16(l-1)+j$
0x800–0xFFF	WEIGHT RAM	RW	2 KiB; byte address $base_l + (\lfloor i/8 \rfloor \cdot 16 + j) \cdot 8 + (i \bmod 8)$, $base_1 = 0$, $base_2 = 1024$
other	reserved	–	read 0, writes ignored

`mem_ready` without `mem_valid`; little-endian byte lanes. Implemented latencies (SIMULATION): RAM, peripherals and the unmapped responder answer in the 2nd cycle, the TPU in the 3rd; a TPU read of INPUT/BIAS/WEIGHT during a tile load or post-processing waits up to 35 cycles (ADR-012 bound 40). There is no bus timeout.

4.4 Top-level ports and pins

Table 7: Top-level ports of `soc_top` and their placement in the routed design (SOURCE: CST with cited sources; SIMULATION: P&R pin report).

Port	Dir	Package pin	IO standard	Note
`clk`	in	52	LVCMOS33	27 MHz oscillator
`btn_rst_n`	in	3	LVCMOS18, pull-up	S1 = pin 3 is an ASSUMPTION
`led_n[0..5]`	out	10, 11, 13, 14, 15, 16	LVCMOS18	active-low LEDs
`uart_tx`	out	17	LVCMOS33	to the on-board USB-UART bridge
`uart_rx`	in	18	LVCMOS33	unused (kept)

5 Analyses and their assumptions

5.1 Resource budget versus result

The pre-RTL budget (`requirements/BUDGETS.md` sec. 1) was built bottom-up on a calibration bench of one 16×8 tile (2855 LUT4 / 1454 FF in modes 0/1, 2967 / 1678 in mode 2, SIMULATION of P&R). It predicted 5735 LUT4 (66.4 %) and 2489 FF for the rev A baseline (mode 1). Table 8 compares it with the release build.

Lesson (risk R-16): block-level budgets were off by +20 % and −51 % and matched in total only by compensation; any future change should be re-budgeted from per-module synthesis of the RTL, using the achieved cost of about 14.4 LUT4 per MAC lane rather than the bench value of 22.

5.2 Latency

Budget model (CALCULATION with ASSUMPTION inputs): $O_{tile} = 36$ load cycles per tile, 16+2 post-processing cycles per layer, 10 argmax and 3 start/done cycles $\Rightarrow N(K) = 10 \cdot 2^K + 409$. RTL (CALCULATION from the FSM, confirmed by SIMULATION for every K and vector): load 33 cycles (32 BSRAM words + 1; input words read in parallel), 19 post-processing cycles per layer, argmax on the fly, CYCLES counts BUSY edges only $\Rightarrow \text{CYCLES}(K) = 10(2^K + 33) + 2 \cdot 19 = 10 \cdot 2^K + 368$. REQ-TIM-004 allows $1.10\,N(K)$ (3265 at $K = 8$); the margin at $K = 8$ is 337 cycles.

Table 8: Budget (CALCULATION) versus achieved (SIMULATION).

Item	Budget	Achieved	Deviation	Explanation
SoC LUT4	5735	5628	−1.9 %	right by compensation: TPU +20 %, CPU −51 %
SoC LUT4 (budget re-evaluated for mode 2 + BSRAM regfile)	5591	5628	+0.7 %	deltas of BUDGETS 1.5 applied
SoC DFF	2489 (2713 mode 2)	2803	+12.6 / +3.3 %	mode 2 chosen after the budget (+224 FF)
SoC BSRAM	3	5	+2 blocks	CPU register file in BSRAM (option taken)
RAM16SDP4	56	16	−33 %	SCORE kept in flip-flops
Worst-seed Fmax	52.12 MHz	47.46 MHz	−8.9 %	critical path moved from the SC MAC to the CPU
TPU latency	$10 \cdot 2^K + 409$	$10 \cdot 2^K + 368$	fixed term −10.0 %	input load overlapped, argmax folded in

Table 9: Latency per inference at 27 MHz: budget (CALCULATION) and RTL (SIMULATION; time and rate are CALCULATION from the cycle count).

K	L	budget $N(K)$	RTL CYCLES	time [µs]	inferences/s	bound $1.1N$
4	16	569	528	19.56	51 136	625
5	32	729	688	25.48	39 244	801
6	64	1049	1008	37.33	26 786	1153
7	128	1689	1648	61.04	16 383	1857
8	256	2969	2928	108.44	9 221	3265
9	512	5529	5488	203.26	4 920	6081
10	1024	10649	10608	392.89	2 545	11713

CPU side (SIMULATION, `logs/soc/tb_soc_demo.log`): START store to RESULT load 2948 cycles (CYCLES + 20); reset release to TOHOST 5441 cycles (201.5 µs), 981 instructions, CPI 5.51. The demo does not write the 64 input codes (they come from `input.hex`), so the CPU-side input-load time of REQ-TIM-005 was not simulated (accepted for V1, ADR-011). Double-buffered weight registers would save 9.7 % at $K = 8$ for +1024 FF (CALCULATION, not implemented).

5.3 Timing

The timing budget assumed the SC MAC path (LFSR → comparator → popcount → accumulator) to be critical at 52.12 MHz and set a rule of at most 7 LUT levels between registers. In the release SoC the critical path is in the CPU (21.07 ns: 8.83 ns logic, 12.24 ns routing; instruction register → decoder/FSM next state → ALU operand select → 32-bit carry chain → PC), 92 % longer than the PicoRV32 proxy used in the budget; the TPU alone reaches 60.43 MHz. The slack against the 33.67 ns budget period is 12.6 ns. The nextpnr Gowin timing model is reverse-engineered and not validated against silicon (risk R-05); this is why the 10 % margin exists.

5.4 Power (estimate only)

No open-source power model exists for GW1N and the vendor estimator is excluded by the open-toolchain requirement. The order-of-magnitude estimate $P_{dyn} = \sum \alpha C V^2 f$ with $V = 1.2$ V and $f = 27$ MHz and the achieved cell counts gives logic 1.1–13.1 mW, clock tree 1.1–5.4 mW, BSRAM 2.5–10 mW and static power of a few to about 20 mW, i.e. about 5–50 mW for the FPGA core (ASSUMPTION: activity and capacitance coefficients and the core voltage are not sourced). Board-level power is dominated by the USB bridge, regulators and LEDs and is not estimated. No power value is presented as SIMULATION or MEASUREMENT.

5.5 UART divider

$27 \times 10^6/115200 = 234.375$; the divider 234 gives 115 384.6 Bd, an error of +0.16 % in rate (−0.16 % in bit time), well inside the 2 % tolerance (CALCULATION); the SoC testbench confirms 234 clocks per bit and 2340 per frame (SIMULATION).

5.6 Accuracy study

Set-up. Bit-exact golden model, 360-image stratified test split (seed 0), weights `qat_noise` with $s = 0$ (selected on the 288-image validation split, then re-trained on all 1437 training images; not re-trained for rev B or C), $K = 4..10$, three RNG modes, deployed seeds and a 20-offset ensemble (all generators advanced by a common random start offset, numpy seed 0, which preserves the relative lags of the deployed set). Accuracies carry the correct count k and a 95 % Wilson interval; paired comparisons use the exact McNemar test (SIMULATION).

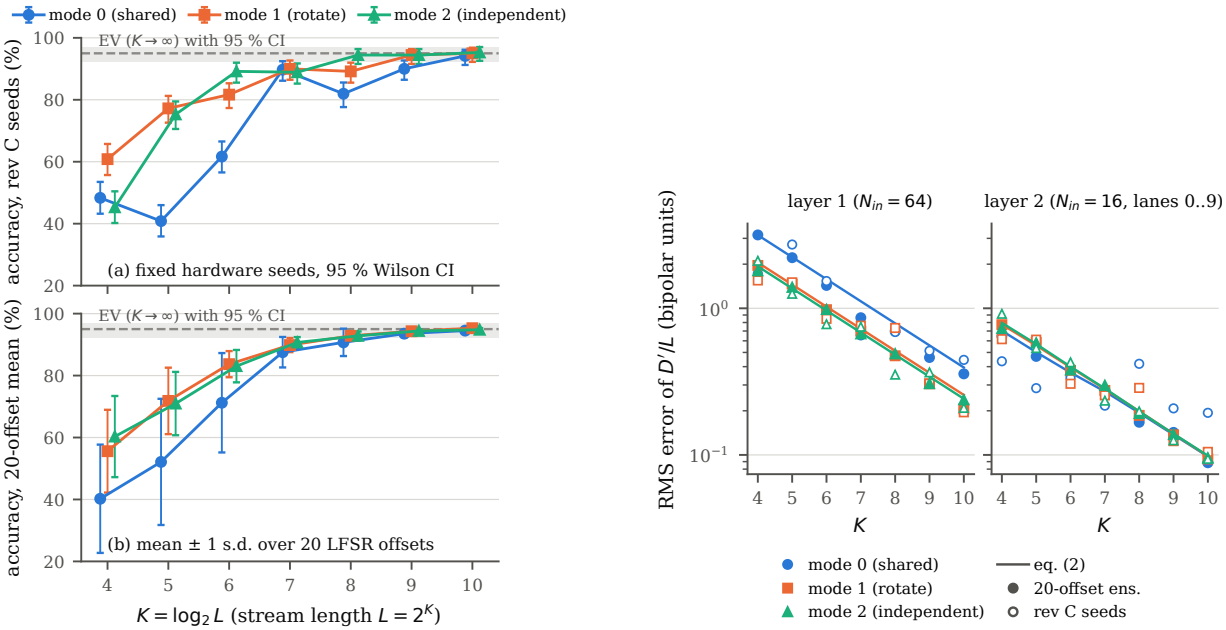


Figure 1: Left: test accuracy versus K for the three RNG modes, deployed rev C seeds with Wilson CI (top) and 20-offset ensemble mean ± s.d. (bottom). Right: RMS stochastic error of D'/L per neuron, ensemble (filled), eq. (1) (lines) and deployed seeds (open). All SIMULATION; data `reports/sw/accuracy_revC.json`.

Table 10: Test accuracy [%] versus K, rev C (SIMULATION, $n = 360$): deployed seeds (k) and ensemble mean ± s.d. over 20 offsets. EV 95.0 (342/360, CI 92.2–96.8). Full CIs in Table II of the article and in the evidence file.

K	m0 HW (k)	m0 ens.	m1 HW (k)	m1 ens.	m2 HW (k)	m2 ens.
4	48.3 (174)	40.2±17.5	60.8 (219)	55.6±13.4	45.3 (163)	60.3±13.1
5	40.8 (147)	52.1±20.4	77.2 (278)	71.8±10.7	75.3 (271)	71.0±10.2
6	61.7 (222)	71.2±16.0	81.7 (294)	83.7±4.2	89.2 (321)	83.0±5.2
7	89.7 (323)	87.5±4.9	90.0 (324)	89.8±2.3	88.9 (320)	90.7±1.7
8	81.9 (295)	90.7±4.4	89.2 (321)	92.8±1.3	94.4 (340)	92.8±1.4
9	90.0 (324)	93.5±1.1	94.4 (340)	94.3±0.8	94.4 (340)	94.4±0.5
10	94.2 (339)	94.5±0.7	95.0 (342)	95.3±0.5	95.3 (343)	94.8±0.5

Findings. (1) At $K = 8$ the deployed seeds give 81.9/89.2/94.4 % (modes 0/1/2) but the ensemble means are 90.7/92.8/92.8 %: a large part of the spread between modes at one seed set is realisation noise. (2) Mode 0 is worse in mean and spread (s.d. 4.4 pp at $K = 8$) and non-monotonic in K. (3) The mode-1 deployed value at $K = 8$ is 2.8 s.d. below its ensemble mean, an unfavourable draw after the rev B change of the modes 0/1 W seed. (4) The exact int8 datapath equals the EV model (0 mismatching hidden codes, scores,

results), so EV is the non-SC baseline. (5) Statistical resolution: one image is 0.28 pp; Wilson half-widths are 2.3–3.1 pp between 95 and 90 %, so differences of 1–2 pp are not resolved.

5.7 Error model

With spins $\sigma = 2b - 1$, uniform cycle-independent r and independent X/W generators (assumptions A1–A4 of `reports/sw/error_model.md`),

$$\mathrm{Var}(D'/L) = \frac{1}{L} \sum_{\text{tiles}} \Big[\sum_{i,k} C^x_{ik} C^w_{ik} - \Big(\sum_i v_{x,i} v_{w,i} \Big)^2 \Big], \tag{1}$$

which reduces to $\frac{1}{L} \sum_i (1 - v^2_{x,i} v^2_{w,i})$ for independent lanes (CALCULATION). Over the rev C 20-offset ensemble the measured/predicted RMS ratio is 0.77–1.05 for all 42 layer/K/mode combinations (SIMULATION); mode 2: 0.90–1.05. A single deployed realisation can deviate by a factor of about 2 (mode 0, layer 2, $K = 8$: 0.42 vs 0.19). Limitations: assumption A2 ignores lagged copies between LFSRs on the common sequence (rev C minimum 321 stream cycles, X_0–W_3, coupling only one operand of two products); the model is per layer and does not propagate errors through the clipped ReLU. For rev B, A4 was violated (translated pair streams) and the model under-predicted mode 2 by 30 % and 55 % at $K = 9$ and 10.

5.8 Variant studies with identical inputs

Each variant study changes one factor and keeps everything else identical: the same trained weights, the same 360 test images and the same 20 offsets for accuracy; the same synthesis wrapper, constraint file, tool versions, 27 MHz constraint and placement seeds 1–3 for implementation.

Table 11: RNG-mode variants, IL = 8; rule K: smallest K within 1 pp of EV, deployed seeds (implementation: SIMULATION of synthesis and P&R of the stand-alone TPU wrapper, rev A RTL, mode-2 seeds identical to rev C; accuracy: SIMULATION, rev C, $K = 8$).

Mode	LUT4 nwl/def	DFF	Fmax nwl/def [MHz]	HW acc.	ens. mean±s.d.	rule K
0 shared	4468 / 4539	2445	59.44 / 54.95	81.9	90.7 ± 4.4	10
1 rotate	4461 / 4567	2445	61.41 / 55.26	89.2	92.8 ± 1.3	9
2 independent	4572 / 4692	2669	60.43 / 54.34	94.4	92.8 ± 1.4	8

Mode 2 costs +111 LUT4 (+125 in the default flow) and exactly +224 DFF (14 extra 16-bit LFSRs) over mode 1; Fmax differences are within seed noise. Mode 2 was chosen (ADR-008/010) for the smallest stream-length requirement with the deployed seeds and the smallest seed-to-seed spread at $K = 9$; the ensemble means of modes 1 and 2 at $K = 8$ are equal, so the advantage is modest and rests partly on the deployed realisation.

Table 12: Input-lane variants, mode 1 (SIMULATION; latency for IL 4/16 is CALCULATION only, IL = 8 confirmed by simulation).

IL	LUT4 nwl / def	DFF	Fmax nwl / def [MHz]	CYCLES at $K = 8$	time [µs]
4	3542 / 3693	1901	65.30 / 71.34	5498	203.6
8	4461 / 4567	2445	61.41 / 55.26	2928	108.4
16	6322 / 6699	3533	no legal placement	1643	60.9

IL = 16 at 73–77 % LUT4 (carry-heavy) failed placement on all attempted seeds (3 default, 1 `-nowidelut`; the remaining two were aborted), so the practical placement limit for such designs can be below the 85 % rule.

The differences between qat and qat_noise on the test split are within one or two images and not significant; qat_noise was selected on the validation split before the test split was used.

Synthesis flow. On the SoC, `synth_gowin -nowidelut` uses 502 fewer LUT4 positions than the default wide-LUT mapping (5628 vs 6130), no MUX2_LUT cells (vs 915) and its worst seed is 4.66 MHz faster (47.46 vs 42.80 MHz); it is the release flow (ADR-011).

Table 13: Training variants (SIMULATION, rev C seeds, test split; the selection was made on the validation split).

Variant	s	EV (k)	mode 2, $K = 8$ (k)	mode 1, $K = 8$ (k)	rule K m2 (1 pp)
float (post-training rounding)	1	95.3 (343)	90.8 (327)	74.7 (269)	none ≤ 10
qat	0	95.8 (345)	95.3 (343)	86.4 (311)	8
qat_noise (selected)	0	95.0 (342)	94.4 (340)	89.2 (321)	8

5.9 Stream length $L = 2^K$ as a design rule

The operating point is the smallest K at which the accuracy stays within a tolerance of EV for every larger K (Table 14). With the deployed seeds mode 2 reaches it at $K = 8$ for both 1 and 2 pp; on the ensemble mean $K = 9$. REQ-AI-005 uses the 2 pp threshold: $K_{op} = 8$, with gaps 0.56, 0.56 and -0.28 pp at $K = 8, 9, 10$. Because 1 pp is 3.6 images and below the CI half-width, the rule is an engineering choice, not a statistical claim. Latency doubles with every step of K (Table 9), so $K = 8$ trades 108 µs per inference against ≤ 1 pp of accuracy.

Table 14: Smallest K within 1 pp | 2 pp of EV for all larger K (SIMULATION).

Rev	Basis	mode 0	mode 1	mode 2
A	deployed	10 | 10	9 | 9	8 | 8
B	deployed	10 | 10	9 | 9	10 | 10
C	deployed	10 | 10	9 | 9	8 | 8
C	ensemble mean	10 | 9	9 | 9	9 | 9

5.10 Seed-rule study (rev A → B → C)

Rev B and its defect. Rev B set $X_i = S(4096i)$, $W_i = S(32768 + 4096i)$ ($S(n)$: LFSR state n single steps after `0xACE1`), maximising the pairwise phase distance (all 16 generators ≥ 512 stream cycles apart). Because X and W advance by the same 4096 steps per lane, lane i's (r_x, r_w) stream is lane $i-1$'s delayed by 512 stream cycles, a whole number of tiles for $K \leq 9$; identical (tile, lane) count tables recur (14/54/120/196/100 pairs of 3160 at $K = 6..10$, rev A: 0) and their errors add coherently. Mode 2 at $K = 9$ dropped to 89.7 % (323/360) from 94.4 % (340/360): discordant 5/22, exact McNemar $p = 0.0015$, difference +4.7 pp (Newcombe 95 % CI +2.0 to +7.8). At $K = 8$ the defect was invisible (93.9 vs 94.4 %, $p = 0.79$).

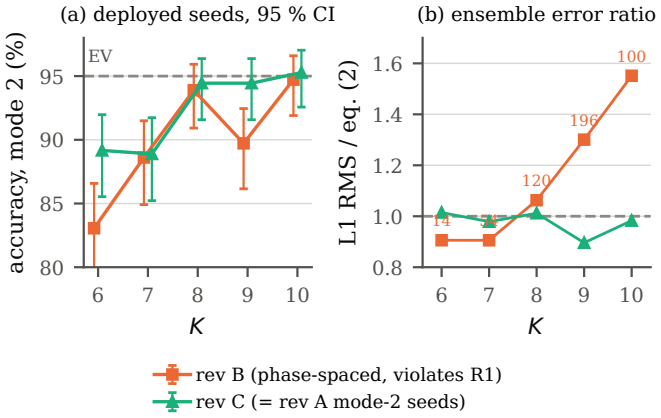


Figure 2: Mode-2 seed iteration, rev B versus rev C (SIMULATION, $n = 360$): (a) deployed-seed accuracy with Wilson CIs; (b) layer-1 ensemble RMS relative to eq. (1), with the identical count-table pairs of rev B.

Rules. With D_i the phase offset in single steps between W_i and X_i: **R1** no two lanes' operand-pair streams are translates ($D_i \neq D_j$, $i \neq j$); **R2** no direct or swapped shared-bit lag between operand pairs ($|D_i - D_j| > 14$ for $i \neq j$ and $|D_i + D_j| > 14$ for all i, j, circular mod 65535; ±14 because r is an 8-bit window of one bit sequence). R2 is stricter; within the rev B lattice it is unsatisfiable. Two independent

implementations (software model and budget script) agree: rev C satisfies R1 and R2 (minimum pairwise lag 321 stream cycles, next 541), rev B violates R1 on 28 of 28 lane pairs and R2 on 28 direct + 36 swapped pairs. The modes 0/1 pair `0xACE1`/`0xF9DE` violates R2 by self-swap (accepted: non-default study modes, no measured effect; ADR-010 D-B).

Table 15: Rev C seed candidates on the 288 validation images (SIMULATION; structure CALCULATION). Caveat: these images were in the final training set; absolute values are optimistic, only the ranking is used.

Cand.	Rule	R1	R2	λ_8 [cyc.]	$K = 8/9/10$ (k/288)	worst [%]	rank
C0	rev A: $X_i =$ `0xACE1` $+$ `0x3C5A`i, $W_i =$ `0x7A5C` $+$ `0x2B67`i	yes	yes	321	285/285/286	98.96	1
C1	rev B: $S(4096i)$, $S(32768 + 4096i)$	no	no	512	279/274/282	95.14	5
C2	rev B, W lanes reversed	yes	no	512	276/282/283	95.83	4
C3	$S(32760i)$, $S(32760(15 - i))$ (max λ_8)	yes	yes	4095	283/284/287	98.26	2
C4	$S(4095i)$, $S(4095(15 - i))$ (max λ_1)	yes	yes	3585	280/284/285	97.22	3

Protocol and outcome. The selection rule was fixed before any accuracy was computed (ADR-009): highest worst-case deployed validation accuracy over $K \in \{8, 9, 10\}$, tie-break on the ensemble mean; C3 and C4 were designed by an exhaustive scan (65 288 feasible step sizes of 65 534) from CALCULATION alone. C0 won by 2 images over C3; among the rule-compliant candidates no difference is significant (paired McNemar $p \geq 0.125$), while the two rule-violating candidates rank last. Rev C thus restores the rev A mode-2 seeds, now justified by the rules and by held-out data, and keeps the longest RTL verification history. The test split was evaluated once afterwards: rev C equals rev A in mode 2 image by image (identity asserted bit-exactly) and rev B in modes 0/1. Lesson: maximal phase spacing alone is not a sufficient criterion; the issue-1 risk mitigation that proposed it was wrong (R-02).

6 Timing and utilisation (three seeds)

Flow: Yosys 0.70 (YoWASP) `synth_gowin -nowidelut`, nextpnr-0.11.1 himbaechel-gowin, device GW1NR-LV9QN88PC6/I5, family GW1N-9C, CST `design/constraints/tangnano9k.cst`, `-freq 27`, `-threads 1`, no `-timing-allow-fail`; Apicula 0.32 `gowin_pack`. All values are tool estimates (SIMULATION).

Table 16: Release SoC utilisation (identical on all seeds; SIMULATION).

Resource	used	device	%	≤85 %
LUT4 (incl. ALU and LUT-RAM positions)	5628	8640	65	PASS
ALU	1976	6480	30	PASS
DFF	2803	6480	43	PASS
BSRAM	5	26	19	PASS
RAM16SDP4	16	270	5	PASS
IOB	10	276	3	PASS
MUX2_LUT5 / MULT18X18	0 / 0	4320 / 20	0	PASS

(PASS in this table refers to the 85 % tool-estimate criterion only; it is not a requirement status.)

Table 17: Post-route Fmax per seed (SIMULATION, static timing of nextpnr; target ≥ 29.7 MHz).

Design	seed 1	seed 2	seed 3	worst	margin vs 29.7 MHz
SoC, `-nowidelut` (release)	51.98	47.46	51.21	47.46	+59.8 %
SoC, default flow	46.47	45.79	42.80	42.80	+44.1 %
CPU + RAM alone	62.46	53.26	55.74	53.26	+79 %
TPU alone, IL 8 mode 2, `-nowidelut`	64.28	64.24	60.43	60.43	+103 %

Critical paths (worst seed): SoC 21.07 ns (CPU instruction register to PC through the decoder, FSM next state and ALU carry chain); default-flow SoC 23.36 ns, same path class; TPU alone (`-nowidelut`) the SC datapath itself (LFSR or weight register → comparator carry chain → XNOR → popcount → 17-bit accumulator); TPU alone, default flow (IL 8, mode 1), the 34-bit post-processing shifter. BSRAM mapping

was checked in the routed netlist: 5 blocks, 3 with non-zero initialisation (program RAM, weight RAM), 2 with all-zero initialisation (register file). All 10 ports sit on their CST pins. The bitstream (3 556 680 bytes, sha256 `152d3c6d...76258`) is a build product in `sim/soc/build/`, regenerated by `scripts/synth_soc.sh`.

7 Verification methodology and results per block

All testbenches are self-checking (print `TEST PASSED` or `TEST FAILED`, `$fatal` on failure, timeout watchdog) and run with Icarus Verilog 12.0; RTL lint with Verilator 5.020 `-Wall`. Each testbench was also run with an injected fault and had to fail.

Table 18: Verification per block: what was checked and the decisive log line (SIMULATION).

Block / TB	Method and volume	Decisive result	Log
Python golden model	two implementations (reference loops / vectorised count tables); 158 unit tests; reference re-generation of every exported vector	`158 passed`; `CROSSCHECK TOTAL: 593 vectors x 33 fields, mismatches = 0`	`logs/sw/pytest_revC.log`, `gen_hex.log`
Python test mutation	11 mutants of the rev C tests (wrong tables, rules, PARAMS, headers, selection)	11/11 detected	`logs/sw/mutation_revC.log`
Assembler + ISS	unit tests incl. round trip of 48 502 instruction words	0 mismatches	`logs/sw/pytest_rv32.log`
CPU (`tb_rv32i_core`)	44 ISS-checked programs (11 directed, 20 random, 13 trap) × 4 bus timings; protocol checker P1–P7; signature, trap cause, halt PC, instret, TOHOST, LED, UART compared	`# runs:  176 / 176 PASS`	`logs/cpu/sim_cpu_summary.log`
LFSR unit (`tb_sc_lfsr8`)	leap vs 8 single steps, 70 000 cycles; period; rev C seed tables re-derived (TC7)	`TC2 LEAP==8x1 70000 cycles, 0 mismatches`; TC7 PASS	`logs/tpu/tb_sc_lfsr8.log`
Post-processing unit	53 328 vectors incl. every K, s, extreme acc/bias	`mismatches = 0`	`logs/tpu/tb_sc_postproc.log`
Lane array unit	9216 stream cycles, forced r at the code boundaries × all 256 codes	`TEST PASSED`	`logs/tpu/tb_sc_lane_array.log`
TPU system (`tb_sc_tpu`)	directed D1–D12 (reset values, PARAMS, clamp, read-back, BUSY behaviour, CYCLES, reset during inference, init files); golden vectors per mode: 360 images, 51 corner, 63 sweep (+32 mode-1 $K = 4/10$); RNG trace per cycle; protocol checker	1454 inferences + 239 clamp re-runs, `mismatching fields 0`; RNG trace 0 mismatches; PARAMS 0x03020810	`logs/tpu/sim_tpu_console.log`
SoC system (`tb_soc`)	pins-only TB: reset synchroniser, protocol checker, real-time UART receiver and bit-timing checker, demo D1–D7, 44 CPU programs on the SoC	demo D1–D7 PASS (TOHOST cycle 5441, RESULT 7, "7\n"); `44 / 44 PASS`	`logs/soc/tb_soc_demo.log`, `cpu_suite_summary.log`
SoC fabric (`tb_soc_bfm`)	CPU outputs forced: register access, byte lanes, memory-map edges, unmapped access, CYCLE, UART busy	T1–T10 PASS	`logs/soc/tb_soc_bfm.log`
Lint	13 TPU configurations, 5 SoC configurations, CPU	`LINT_OK`, no warning	`logs/tpu/lint_rtl.log`, `logs/soc/lint_rtl.log`, `logs/cpu/lint.log`

These faults target the checkers, references and stimuli; a systematic mutation campaign of the RTL itself was not run.

Not run. Gate-level (post-synthesis or post-route) simulation: **NOT RUN**, because the flow provides no behavioural models of the Gowin block RAMs. Hardware test: **NOT RUN** (out of scope; Section 11).

Table 19: Negative runs (each must end in `TEST FAILED` with a non-zero exit; SIMULATION).

Testbench	Injected fault	Result
`tb_sc_lfsr8`	wrong feedback tap in the reference	TEST FAILED
`tb_sc_postproc`	truncating instead of floor division in the reference	TEST FAILED
`tb_sc_lane_array`	AND instead of XNOR in the reference	TEST FAILED
`tb_sc_tpu`	bit 0 of SCORE[0] flipped in the comparison	TEST FAILED (16 mismatching inferences)
`tb_rv32i_core`	corrupted expected signature; request changed while waiting	TEST FAILED, `VVP_EXIT=1` (both)
`tb_soc`	expected SCORE[0] bit flipped; UART divider 240; `mem_ready` without `mem_valid`	TEST FAILED (all three)
`tb_soc_bfm`	LED polarity expected non-inverted	TEST FAILED

8 Review findings

Pending. The independent review REV-01 (`reports/review/REVIEW_A_cpu_soc.md` for CPU and SoC, `reports/review/REVIEW_B_tpu_model_article.md` for TPU, golden model and article) was running when this report was written. Its findings, their severity and their disposition will be summarised here when they are available. Nothing in this report has been reviewed independently yet.

9 Risks

Table 20: Risk register summary (`management/RISKS.md` issue 2).

ID	Risk	State	Note
R-01	SC accuracy too low	MITIGATED	mode 2 K8 94.4 %; CI lower bound 1.6 pp above the 90 % threshold
R-02	LFSR correlation / seeds	REALISED (rev B), MITIGATED (rev C)	R1/R2 + pre-registered selection; modes 0/1 pair violates R2 (accepted)
R-03	Resource overflow	RETIRED (frozen design)	65 % LUT4
R-04	Timing closure	MITIGATED	47.46 MHz worst seed
R-05	Open-toolchain maturity	OPEN	no gate-level simulation; timing model and bitstream unvalidated on silicon; IL 16 placement failure below 85 %
R-06	Model/RTL bit-exactness drift	MITIGATED (simulation level)	0 mismatching fields; gate level not run
R-07	Unverifiable data-sheet values	OPEN	vendor hosts blocked; user decision OD-16
R-08	No hardware validation	OPEN (by scope)	plan in Section 11
R-09	Pin ambiguity S1/S2	PARTIAL	S1 = pin 3 still an assumption (PV-05)
R-10	Statistical over-claiming	OPEN	CIs and McNemar used; wording to be checked by the reviewer
R-11	Latency above budget	RETIRED	$2928 < 3265$
R-12	Buffer port sharing	OPEN	stalled read up to 35 cycles (ADR-012 bound 40)
R-13	Repository bloat	MITIGATED	build products ignored
R-14	Optimistic seed selection	OPEN	validation images seen in training; C0 vs C3 not significant
R-15	Revision mix-up A/B/C	OPEN	spec-revision guards in data and scripts
R-16	Budget-method weakness	OPEN	block budgets +20 % / −51 %

10 Limitations

- **Simulation only.** No quantity is a hardware measurement; the bitstream was never loaded. Fmax comes from a reverse-engineered, not silicon-validated timing model; there are no I/O timing constraints (all external interfaces are slow or asynchronous).
- **No gate-level simulation** (no behavioural BSRAM models): the equivalence of the netlist with the RTL rests on the synthesis tool.
- **CPU-side latency incomplete** (REQ-TIM-005 PARTIAL): the demo pre-loads the input image; writing 64 codes by the CPU was not simulated.
- **Accuracy resolution**: $n = 360$; differences below about 3 pp are not significant without paired tests; the seed selection used validation images that the final weights had seen.
- **Workload**: a 64-16-10 MLP on 8×8 digits; results may not transfer to larger networks. The network was not re-trained for rev B/C.
- **Throughput**: 1.09×10^7 useful MAC/s at $K = 8$ against a theoretical 1.08×10^9 int8 MAC/s with the hard multipliers (CALCULATION; no int8 design built).
- **Sources**: Gowin data sheet and memory guide not accessible; BSRAM organisation, core voltage and the S1/S2 button-to-pin mapping are ASSUMPTIONs.
- **No power figure** beyond the order-of-magnitude ASSUMPTION.
- **UART RX unused**: a host cannot send data to the SoC in rev C, which limits what a hardware test can do without a design change.
- **Single LFSR polynomial**: all generators are lagged copies of one sequence; at $K = 10$ each generator re-uses its bit sequence shifted by one bit after 8192 stream cycles (no measured penalty).

11 Physical validation plan

This plan satisfies the documentation part of REQ-VER-009. **None of it was executed in this project**; every step has the status **NOT RUN**. Loading a bitstream into a board is a decision of the user (see the user manual, `reports/engineering/MANUAL.md`). Results obtained by following this plan are the first MEASUREMENTs of the project and must be recorded with instrument, setting, date, board serial and the git revision and sha256 of the bitstream used.

11.1 Equipment

- Sipeed Tang Nano 9K and a USB-C data cable; host PC with `openFPGALoader` (not part of the project tool set) and a serial terminal (e.g. 115200 8N1, no flow control).
- USB-UART: the on-board USB bridge of the Tang Nano 9K exposes the FPGA UART (pins 17/18) as a serial port (SOURCE: CST sources [A]/[B]).
- Logic analyser (≥4 channels, ≥10 MS/s) or oscilloscope; probe access to pin 17 (UART TX), the reset button signal (pin 3) and one LED pin.
- Bench supply (5.0 V, current limit 0.5 A) feeding the board through a USB-C power breakout, with a current meter (resolution ≤1 mA), or an in-line USB power meter.
- Optional: thermal camera.

11.2 Preconditions

1. Rebuild the bitstream from a clean checkout of the release revision with `scripts/synth_soc.sh`; record its sha256 and compare with `152d3c6d...76258` (rev C, `logs/soc/pack.log`). A different hash after a rebuild is not necessarily an error (record it), but then the test result refers to the new file.
2. Re-run `scripts/sim_soc.sh` on the same revision and keep the log: the hardware result is compared with that simulation.
3. Load the bitstream into the volatile SRAM configuration only (no flash write), so a power cycle restores the previous board content.

11.3 Test items

Table 21: Physical validation items. Expected values are SIMULATION results of rev C unless labelled otherwise; status of every item: NOT RUN.

Item	What	How (equipment, procedure)	Pass criterion	Req.
PV-01	Supply current, unconfigured and configured	Bench supply 5.0 V via USB-C breakout and current meter (or USB power meter). Record current (a) with the factory/previous configuration, (b) after loading the rev C bitstream, idle after the demo (the program ends in a loop). Repeat for three power-ups.	Values recorded with the measurement label; no current limit trip; no pass bound exists (REQ-SYS-017 is an estimate: FPGA core 5–50 mW, ASSUMPTION, board dominated by the USB bridge).	SYS-017, VER-009(5)
PV-02	3.3 V rail (optional)	Multimeter on the 3.3 V regulator output; the test point must be taken from the official schematic, which was not accessible here (ASSUMPTION).	Recorded; within the regulator tolerance stated in the schematic/data sheet	VER-009
PV-03	Configuration (SRAM load)	`openFPGALoader -b tangnano9k <file>.fs` (see manual); observe the loader's result.	Loader reports success; the design starts (PV-04).	SYS-001, VER-005
PV-04	LED pattern after reset	Hold the reset button (pin 3), then release; observe the six LEDs.	While held: all six off (`led_n`=111111). After release: LEDs on pins 10, 11, 13 lit, 14, 15, 16 off (LED = RESULT = 7; `led_n`=111000).	SYS-006, SYS-013, SYS-016
PV-05	Reset button behaviour and S1/S2 pin assumption	Open the terminal, press and release each button in turn; logic analyser on pin 3 to see bounce.	Exactly one button (expected: silkscreen S1) restarts the demo and the terminal shows a new "7\n"; the other has no effect. Record which label is pin 3; if it is S2, correct the CST comment and SOURCES S-022. Bounce may cause several restarts per press (no debouncer): record their number; repeated "7" lines are acceptable.	SYS-005, SYS-014, R-09
PV-06	UART content	Serial terminal at 115200 8N1 on the board's USB serial port; terminal opened before the reset is released.	Received bytes 0x37 0x0A ("7\n") once per reset; no framing errors.	SYS-012, SYS-016
PV-07	UART timing / clock	Logic analyser on pin 17 during the transmission; measure the duration of the 10-bit frame of 0x37 and of single bits.	Bit time 8.667 µs, frame 86.67 µs (CALCULATION: 234 / 2340 clocks at 27 MHz); baud within 115200 ±2 % (bit time 8.51–8.86 µs). The measured bit time also gives the oscillator frequency $f = 234/t_{bit}$.	SYS-012, SYS-004
PV-08	Start-up latency	Logic analyser: reset release edge (pin 3) to the falling edge of the first UART start bit.	Compare with the same interval in the SoC simulation (to be extracted from the `tb_soc` waveform; not yet extracted). Agreement within a few clock cycles plus the synchroniser latency is expected (deterministic design).	SYS-005, TIM-005

Item	What	How (equipment, procedure)	Pass criterion	Req.
PV-09	Inference latency (CYCLES)	The demo stores CYCLES at RAM `0x0E00` but does not transmit it. A future test program (generated with the Python assembler, not firmware) must read CYCLES and CYCLE and send them as hexadecimal text over the UART.	CYCLES = 2928 exactly at $K = 8$ (and $10 \cdot 2^K + 368$ for other K): the counter is deterministic, so any deviation is a failure.	TPU-014, TIM-004, VER-009(4)
PV-10	RESULT for several images	Requires either (a) ≥20 bitstreams with different `input.hex`, or (b) a future test program that loops over ≥20 images stored in RAM (20 × 64 B fits in 4 KiB) and prints each RESULT.	Each RESULT equals the golden-model RESULT of the same image, mode 2, $K = 8$, rev C (bit-exact expectation: 100 % agreement, not an accuracy threshold).	TPU-013, VER-009(3)
PV-11	Accuracy over the test set	Streaming the 360 test images from the host needs a UART receiver in the RTL (unused in rev C), a test program and a future host script. Not possible with the rev C bitstream.	Future work: hardware accuracy equals the simulated 94.4 % (340/360) image by image.	AI-004 (future)
PV-12	Repeatability	Repeat PV-04, PV-06, PV-07 after three separate power-ups.	Identical results each time.	VER-009(6)
PV-13	Fmax margin by PLL sweep	Future work: a separate test build with a PLL (excluded from rev C by REQ-SYS-004) sweeping the system clock upward in steps, with the UART divider recomputed per frequency; PV-06/PV-10 at each step.	Correct operation at ≥ 29.7 MHz; the highest passing frequency is compared with the 47.46 MHz STA estimate.	TIM-001 (future)
PV-14	Thermal (optional)	Thermal camera on the FPGA package after 10 min in the demo loop.	Informative only; no hot spot on the board above normal ambient-plus-a-few-kelvin (ASSUMPTION, no requirement).	–

11.4 Order and stop criteria

Run PV-01 (unconfigured), PV-03, then PV-04 to PV-07, PV-12 and PV-01 (configured); PV-08 to PV-11 and PV-13 need additional work. Stop and disconnect if the supply current trips the limit, the board heats noticeably or the loader reports an error. Any failure is recorded as a hardware **FAIL** with the evidence and does not invalidate the simulation results, which remain labelled as such.

12 Reproduction

The commands to reproduce every result, the expected decisive output lines, the repository map and troubleshooting are in the user manual `reports/engineering/MANUAL.md`. Main entry points: `scripts/sim_cpu.sh` (∼10 s), `scripts/sim_tpu.sh` (∼45 min), `scripts/sim_soc.sh` (∼2 min), `scripts/synth_soc.sh` (∼45 min single-threaded), and in `code/python`: `python -m sctpu.accuracy -spec-rev C -ensemble-size 20`, `python -m sctpu.seed_select`, `python -m sctpu.compare_abc`.